

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 13

Measurement: How the World Gets Numbers

Exercises

Self-Study Curriculum

1. Classify each of the following by scale type, and state one operation that is *not* meaningful on it: pH; qPCR Ct value; a 5-point Likert symptom score; molar concentration; tumor grade (I-IV); temperature in $^{\circ}\text{C}$; species identity. For any that are genuinely ambiguous, say what extra information about the empirical operations would settle it.

2. A study reports the “mean tumor grade” of two treatment groups and concludes that group A is worse. Using the invariance test, construct an admissible re-labeling of the grades under which the conclusion reverses. Then say what analysis the authors *should* have run, and why it is safe.

3. Log-fold-change is a workhorse in your field. Which scale type is a raw concentration? Which is its logarithm? Explain what the log transform does to the group of admissible transformations, and why “a log-fold-change of 2” is meaningful while “a mean of two pH values” is more delicate than it looks.

4. Give an example from your own work where treating ordinal data as interval was *justified* (per Velleman & Wilkinson) and one where it silently corrupted a conclusion. What distinguishes the two cases?
